

## No. of Functional Features Included in the Solution

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <b>Date</b>          | 03 July 2026                                                 |
| <b>Team ID</b>       | XXXXX                                                        |
| <b>Project Name</b>  | OptiCrop – Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                      |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                      | Feature Description                                                                                                                          | Module / Component   | Status | Marks Contribution |
|------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------|--------------------|
| 1    | Intelligent Crop Recommendation   | Predicts the most suitable crop using soil nutrients (N, P, K) and environmental conditions such as temperature, humidity, pH, and rainfall. | ML Prediction Engine | Done   | Core Feature       |
| 2    | Soil & Climate Data Input         | Accepts user input through an intuitive web interface for accurate crop prediction.                                                          | User Interface       | Done   | High               |
| 3    | Machine Learning Prediction Model | Uses the trained Random Forest model to generate highly accurate crop recommendations.                                                       | Machine Learning     | Done   | Core Feature       |
| 4    | Prediction Confidence Score       | Displays confidence percentage along with the recommended crop to improve user trust.                                                        | Prediction Module    | Done   | Medium             |
| 5    | Agricultural Data Visualization   | Presents crop distribution, soil analysis, and model insights using interactive charts and graphs.                                           | Analytics Dashboard  | Done   | High               |
| 6    | Dataset Preprocessing             | Cleans, validates, and transforms agricultural data before model training and prediction.                                                    | Data Processing      | Done   | High               |

|   |                              |                                                                                                           |                     |      |        |
|---|------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------|------|--------|
| 7 | Model Performance Evaluation | Evaluates the model using Accuracy, Confusion Matrix, and Classification Report for reliable predictions. | Evaluation Module   | Done | Medium |
| 8 | Responsive Web Application   | Provides an easy-to-use interface accessible from desktop and mobile browsers.                            | Frontend (Flask UI) | Done | High   |

### Feature Summary:

| Metric                             | Count / Value                                                                                        |
|------------------------------------|------------------------------------------------------------------------------------------------------|
| Total Features Planned             | 8                                                                                                    |
| Total Features Implemented         | 8                                                                                                    |
| Core / Must-Have Features          | 5(Crop Recommendation, ML Prediction, Data Input, Dataset Processing, Model Evaluation)              |
| Additional / Nice-to-Have Features | Agricultural Insights, Future Feature Support                                                        |
| Features Tested & Verified         | Soil Data Input, Crop Recommendation, Crop Suitability Analysis, ML Prediction, Prediction Dashboard |

### Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                                 |
|------|---------------------------|----------------------|------------------------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | Soil data input form, Prediction result dashboard                |
| 2    | Backend / Logic           | 3                    | Crop recommendation, Crop suitability analysis, Input validation |
| 3    | Database / Storage        | 1                    | Crop dataset (CSV) and trained ML model                          |
| 4    | API / Integration         | 1                    | Flask application connecting frontend with the prediction model  |
| 5    | Security / Authentication | 1                    | Input Validation and Secure Request Handling                     |